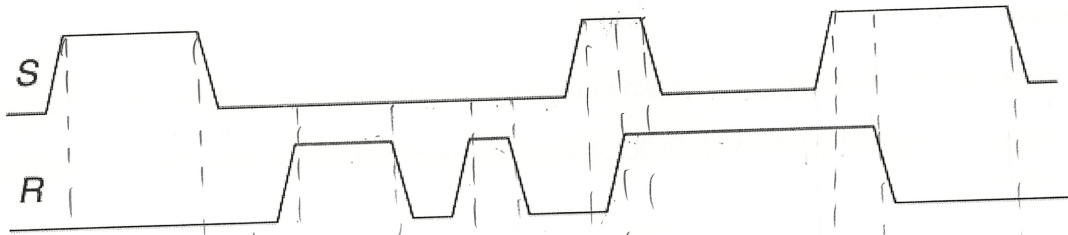


Homework #3

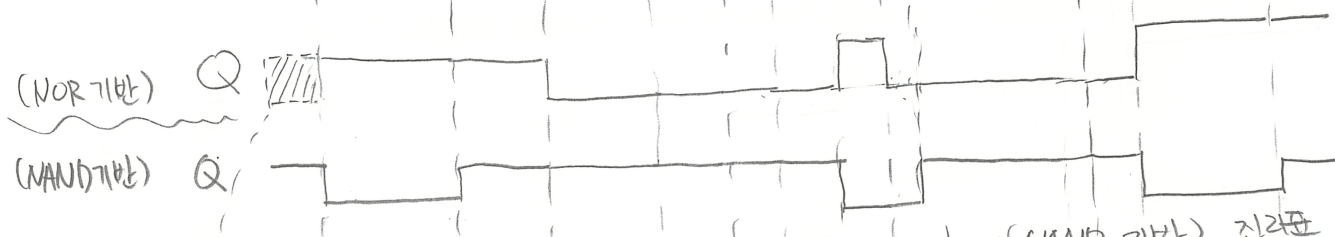
Notice:

- ✓ Solve the problems 1 ~ 7. The problem must be solved by hand. (Use note or blank paper.)
- ✓ Scan and upload your result to KLAS assignment before due date. No late submission is allowed.
- ✓ You should solve the problems yourself. All answers should be written in Korean except any terminologies. No English sentence is allowed.
- ✓ All answers **must have enough solutions** to solve the problem.
- ✓ Deadline – 23:59 on May 31st, 2024, Friday (UTC+9, Korean Standard Time)

1. The input waveform is given as below.



A. Sketch the output Q of an SR Latch.



S R 이 00 일때
이전 Q 를 나타내야 하는데
이전 Q 에 대한 정보가
없음으로 이렇게 표현

* (NOR 기반) SR 래치 진리표

S	R	Q
0	0	이전 Q
0	1	0
1	0	1
1	1	0 출력

Q와 Q' 모두 0으로 내려서
상호관계를 아지 못해
유도하지 않은 입력이다.
그러나 이때 클럭은 안동작
모든 0으로 반응함으로 이 타이밍에서 0으로 출력되도록 하였다.

* (NAND 기반) 진리표

S	R	Q
0	0	1 (유효 X) → 역시 마찬가지이다.
0	1	0
1	0	1
1	1	이전 Q

B. What is the difference between a latch and a flip-flop? Under what circumstances is each one preferable?

래치는 레벨 트리거, 플립플롭은 엣지 트리거이다.

플립플롭은 클럭이 0에서 1로 변할때마다 (or 1에서 0으로) 값이 변하게 된다.

반면 래치는 클럭이 1인 동안 계속해서 그 상태를 유지(경고) 한다. 값을 저장한 기억

(클럭이 HIGH 일때만 값을 출력, 클럭이 LOW 일때는 기입값을 보관해두는 것이 동작)

→ 래치는 한때만도 사용가능

→ 래치: 간단한 타이머 저장, 버퍼링, 스위칭, 등 비동기적 상황에서 유용.

→ 플립플롭: 동기식 타이머, 시퀀스 로직, 정확한 타이밍이 필요한 것들에 유용

* 래치가 유용한 경우: 데이터의 변화가 불확실해, 클럭과 독립적으로 빠르게 반응해야 할 때.

(래치는 클럭이 항상 존재하는 동안 계속해서 입력 데이터의 변화 반영 가능)

* 플립 " : 정확한 클럭 신호를 감지하면 그 시점의 데이터를

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

2. Multiplier is a circuit that multiplies two binary number. For example, if we multiply 1001_2 (9_{10}) and 0100_2 (4_{10}), output should be $0010\ 0100_2$ (36_{10}). Design a 2-bit multiplier as shown in Figure 1.

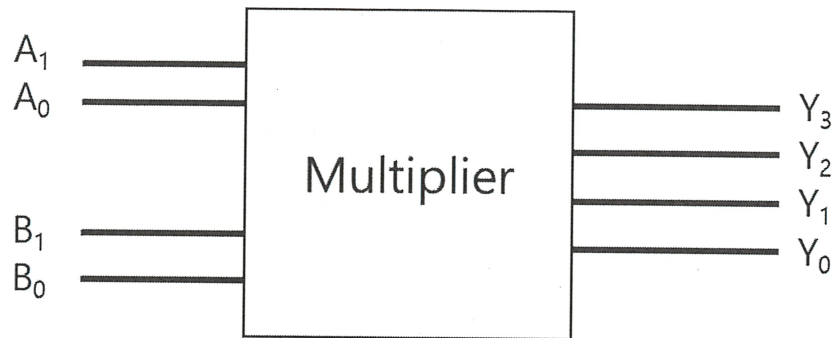


Figure 1 2-bit multiplier

A. Complete a truth table of 2-bit multiplier.

A_1B_0	B_1A_0	C_1	A_1	A_0	B_1	B_0	C_1	Y_3	Y_2	Y_1	Y_0	C_2
0 0 0	0	0	0	0	0	0	0	0	0	0	0	0
0 0 0	0	0	0	0	0	1	0	0	0	0	0	0
0 0 0	0	0	0	0	1	0	0	0	0	0	0	0
0 0 0	0	0	0	0	1	1	0	0	0	0	0	0
0 0 0	0	0	0	1	0	0	0	0	0	0	0	0
0 0 0	0	0	0	1	0	1	0	0	0	0	1	0
0 1 0	0	1	0	1	1	0	0	0	0	1	0	0
0 1 0	0	1	0	1	1	1	0	0	0	1	1	0
0 0 0	1	0	0	0	0	0	0	0	0	0	0	0
1 0 0	1	0	0	0	1	0	0	0	0	1	0	0
0 0 0	1	0	1	0	1	0	1	0	1	0	0	0
1 0 0	1	0	1	0	1	1	0	0	1	1	0	0
0 0 0	1	1	0	0	0	0	0	0	0	0	0	0
1 0 0	1	1	0	0	1	0	0	0	0	1	1	0
0 1 0	1	1	1	0	1	0	1	0	1	1	0	0
1 1 1	1	1	1	1	1	1	1	1	0	0	1	1

$$A = A_1A_0$$

$$B = B_1B_0$$

$$A \times B \Rightarrow \begin{array}{r} A_1A_0 \\ \times B_1B_0 \\ \hline \end{array}$$

$$(C_1) \quad \begin{array}{r} A_1B_0 \quad A_0B_0 \\ A_1B_1 \quad A_0B_1 \end{array} \times$$

$$(C_2) \quad \begin{array}{r} A_1B_1 + A_1B_0 + A_0B_0 \\ C_1 \quad A_0B_1 \\ C_2 \quad C_1 \end{array}$$

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

$$Y_3 \quad Y_2 \quad Y_1 \quad Y_0$$

$$\begin{aligned} Y_0 &= A_0B_0 \\ Y_1 &= A_1B_0 + A_0B_1 \\ Y_2 &= A_1B_1 + C_1 \\ Y_3 &= C_2 \end{aligned}$$

$$\begin{array}{r} 11 \\ 11 \\ 11 \\ \hline 1001 \end{array}$$

B. Boolean equation of Y

 Y_3

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	1	0
10	0	0	0	0

$$Y_3 = A_1A_0B_1B_0$$

 Y_2

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	1
10	0	0	1	1

$$Y_2 = A_1\bar{A}_0B_1 + A_1B_1\bar{B}_0$$

 Y_1

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	0	1	0	1
10	0	1	1	0

$$Y_1 = A_1\bar{B}_1B_0 + A_1\bar{A}_0B_0 + \bar{A}_1A_0B_1 + A_0B_1\bar{B}_0$$

 Y_0

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	0	1	1	0
10	0	0	0	0

$$Y_0 = A_0B_0$$

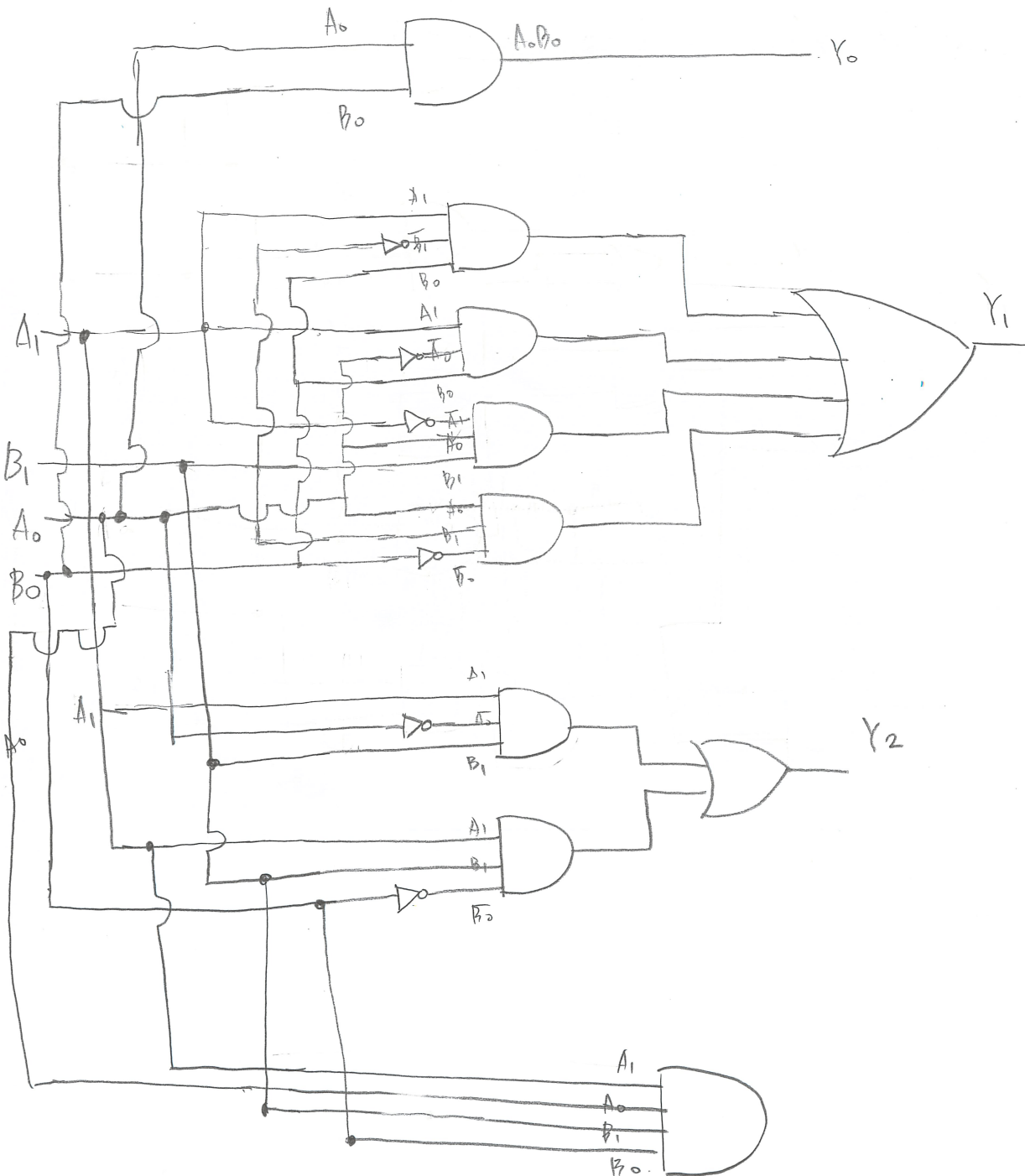
"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

C. Schematic of 2-bit multiplier

$$Y_0 = A_0 B_0$$

$$Y_1 = A_1 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_0 + \bar{A}_1 A_0 B_1 + A_0 B_1 \bar{B}_0$$

$$Y_2 = A_1 \bar{A}_0 B_1 + A_1 B_1 \bar{B}_0, \quad Y_3 = A_1 A_0 B_1 B_0$$

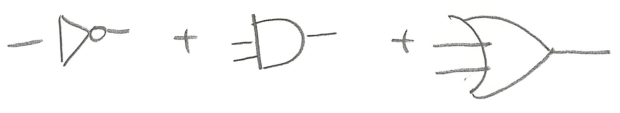


"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

D. In your schematic, which path is critical path? You should explain why.

critical path 는 앞 리로에서 가장 긴, 즉 가장 많은 게이트를
통과하는 길이다. 따라서:

NOT gate + AND gate + OR gate 이다.



3. Find a minimal Boolean equation for the function in table below using K-map and Quine-McCluskey method. Then compare the results of K-map and Quine-McCluskey method. Is it same or not? If not, describe why.

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	X
0	0	1	0	1
0	0	1	1	0
0	1	0	0	X
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	X
1	0	0	1	1
1	0	1	0	X
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	X

<K-map>

②

CD \ AB	00	01	11	10
00	0	X	0	X
01	X	0	1	1
11	0	1	X	0
10	1	1	1	X

$$Y = C\bar{D} + BC + A\bar{C}D$$

<QM>

1248

$$f(A,B,C,D) = \sum m(2,6,7,9,13,14) + \sum d(1,4,8,10,15)$$

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
124 24 3 2 3 3 1 1 1 2 4

		column 1	column 2	column 3
G ₁	1	0001 ✓	-001 ⊗	- - 10
	2	0010 ✓	0-10 ✓	- - 10 ⊗
	4	0100 ✓	-010 ✓	- - 10
	8	1000 ✓	01-0 ⊗	- 11 - ⊗
			100- ⊗	- 11 -
G ₂	6	0110 ✓	10-0 ⊗	- 11 -
	9	1001 ✓	011- ✓	- 11 -
	10	1010 ✓	-110 ✓	- 11 -
G ₃	7	0111 ✓	1-01 ⊗	- 11 -
	13	1101 ✓	1-10 ✓	- 11 -
	14	1110 ✓	-111 ✓	- 11 -
G ₄	15	1111 ✓	111- ✓	- 11 -

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

(This page is left blank. Feel free to use.)

pls

True Minterms	2	6	7	9	13	14
	0010	0110	0111	1001	1101	1110
-001				X		
01-0		X				
100-				X		
10-0						
1-01				X	X	
11-1					X	
-10	(X)	X				
-11-		(X)	(X)			(X)

$Y = A\bar{C}D + C\bar{D} + BC$

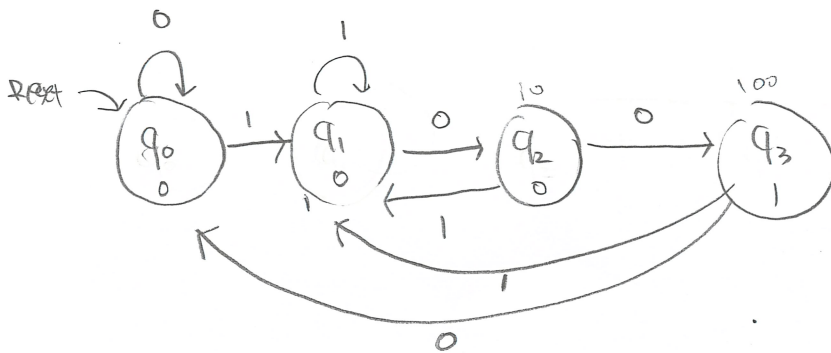
→ K-MAP 과 QM의 결과가 같게 나왔다.

1

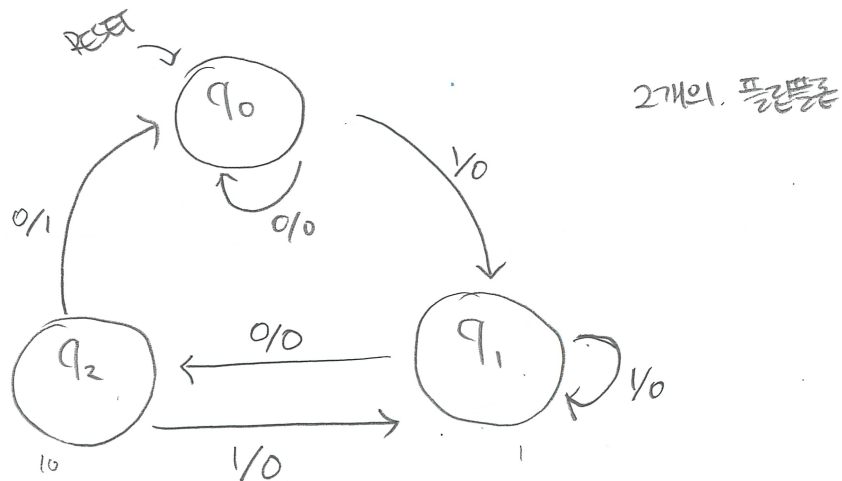
4. Design a sequence detector whose output changes from 0 to 1 when the sequence 100 of an input is detected. For example, if the sequence of 1001100 is given as input, the sequence of output (except initial state) should be 0010001 because there are two 100 sequences.

A. Draw the state transition diagram and find the minimum number of flip-flops in each FSM.

i. In Moore machine



ii. In Mealy machine



B. Draw an encoded state transition table and derive the equation(s) of the flipflop input(s)

i. In Moore machine

State	Encoding
q0	00
q1	01
q2	10
q3	11

	Current State		Input	Next State	
	q ₁	q ₀		q' ₁	q' ₀
q0	0	0	0	0	0
	0	0	1	0	1
	0	1	0	1	0
q1	0	1	1	0	1
	1	0	0	1	1
q2	1	0	1	0	1
	1	1	0	0	0
q3	1	1	1	0	1

q'_1

A \ q ₁ q ₀	00	01	11	10
0	0	1	0	1
1	0	0	0	0

$$\begin{aligned}
 q'_1 &= \bar{A} \bar{q}_1 q_0 + \bar{A} q_1 \bar{q}_0 \\
 &= \bar{A} (\bar{q}_1 q_0 + q_1 \bar{q}_0) \\
 &= \bar{A} (q_0 \oplus q_1)
 \end{aligned}$$

q'_0

A \ q ₁ q ₀	00	01	11	10
0	0	0	0	1
1	1	1	1	1

$$q'_0 = A + q_1 \bar{q}_0$$

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

B. Draw an encoded state transition table and derive the equation(s) of the flipflop input(s)
(cont'd)

ii. In Mealy machine

State	Encoding
q0	00
q1	01
q2	10

Current State	Input	Next state	output
q ₁ q ₀	A	q ₁ ' q ₀ '	Y
q0 0 0	0	0 0	0
0 0	1	0 1	0
q1 0 1	0	1 0	0
0 1	1	0 1	0
q2 1 0	0	0 0	1
1 0	1	0 1	0
x 1 1	0	x	x
1 1	1	x	x

q₁'

A/q ₁ q ₀	00	01	11	10
0	0	1	X	0
1	0	0	X	0

$$q_1' = \bar{A} q_0$$

q₀'

A/q ₁ q ₀	00	01	11	10
0	0	0	X	0
1	1	1	X	1

$$q_0' = A$$

C. Draw an output table and derive the equation of output signal Y

i. In Moore machine

State		Output
$q_1 q_0$		Y
q_0	00	0
q_1	01	0
q_2	10	0
q_3	11	1

$$Y = q_1 q_0$$

ii. In Mealy machine

State	Input	Output
$q_1 q_0$	A	Y
00	0	0
00	1	0
01	0	0
01	1	0
10	0	1
10	1	0
11	X	X
11	X	X

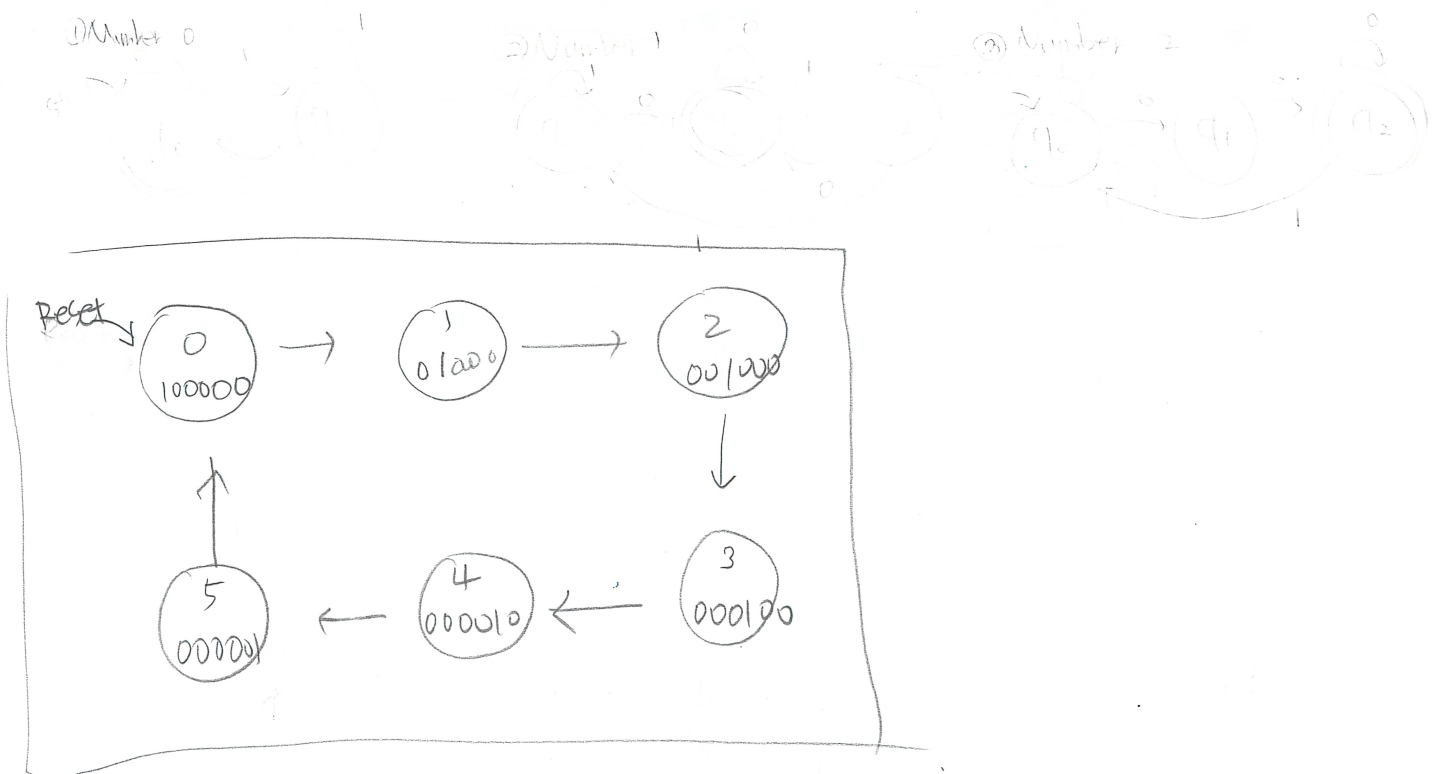
Y	A \ $q_1 q_0$	00 01 11 10			
		0	0	X	1
	0	0	0	X	1
	1	0	0	X	0

$$Y = \bar{A} q_1$$

5. Ring counter, also known as one-hot counter, is a special kind of counter which has one-hot output. Unlike N-bit binary counter, which can count 2^N numbers, N-bit ring counter can count N numbers. For example, outputs of 3-bit ring counter are 100, 010, 001 for 0, 1, 2, respectively. Design a 6-bit ring counter with no input, which starts from 100000 and ends with 000001. On each clock edge, the output should advance to next one-hot. After reaching 000001, it goes back to 100000.

Gate Number	Q ₀	Q ₁	One-hot	Q ₃	Q ₄	Q ₅
0	1	0	0	0	0	0
1	0	1	0	0	0	0
2	0	0	1	0	0	0
3	0	0	0	1	0	0
4	0	0	0	0	1	0
5	0	0	0	0	0	1

- A. Finite state machine with output in each state transition diagram of ring counter with a minimum number of states.



- B. Design a ring counter in finite state machine using states you defined above. You should provide state encoding table, state transition table, output table and Boolean equation of output. Also, you should use **minimum number of flip-flops** and simplify your equation as you can. (No schematic is required)

State	Encoding	Current state	Next state
		$Q_2 Q_1 Q_0$	$D_2 D_1 D_0$
0	000		
1	001		
2	010	0 0 0	0 0 1
3	011	0 0 1	0 1 0
4	100	0 1 0	0 1 1
5	101	0 1 1	1 0 0
		1 0 0	1 0 1
		1 0 1	0 0 0

Current state	Output Y
$Q_2 Q_1 Q_0$	$Y_5 Y_4 Y_3 Y_2 Y_1 Y_0$
0 0 0	1 0 0 0 0 0
0 0 1	0 1 0 0 0 0
0 1 0	0 0 1 0 0 0
0 1 1	0 0 0 1 0 0
1 0 0	0 0 0 0 1 0
1 0 1	0 0 0 0 0 1

Current state	Output
$Q_2 Q_1 Q_0$	$Y_2 Y_1 Y_0$
0 0 0	0 0 1
0 0 1	0 1 0
0 1 0	0 1 1
0 1 1	1 0 0
1 0 0	1 0 1
1 0 1	0 0 0

$Q_2 Q_1$	Q_0	00	01	11	10
0	0	0	1	X	0
1	0	1	X	0	0

$$Y_1 = \overline{Q_2} \overline{Q_1} Q_0 + Q_1 \overline{Q_0}$$

$Q_2 Q_1$	Q_0	00	01	11	10
0	0	0	0	X	1
1	0	0	1	X	0

$$Y_2 = Q_2 Q_0 + Q_2$$

$$Y_0 = \overline{Q_0}$$

$Q_2 Q_1$	Q_0	00	01	11	10
0	0	1	1	X	1
1	0	0	0	X	0

$$Y_5 = \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

$$Y_4 = \overline{Q_2} \overline{Q_1} Q_0$$

$$Y_3 = \overline{Q_2} Q_1 \overline{Q_0}$$

$$Y_2 = \overline{Q_2} Q_1 Q_0$$

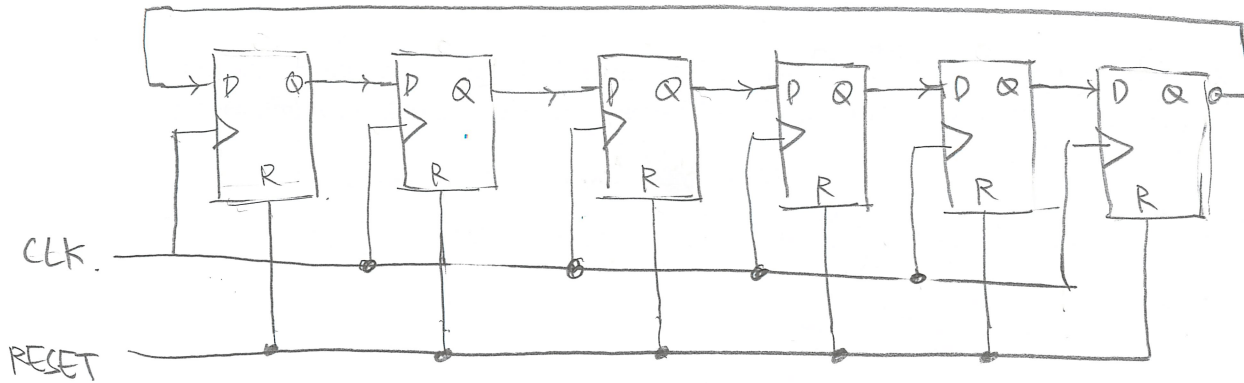
$$Y_1 = Q_2 \overline{Q_1} \overline{Q_0}$$

$$Y_0 = Q_2 \overline{Q_1} Q_0$$

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

C. Is there any other easy way to design a 6-bit ring counter?

먼저 기본 6 bit ring counter 는



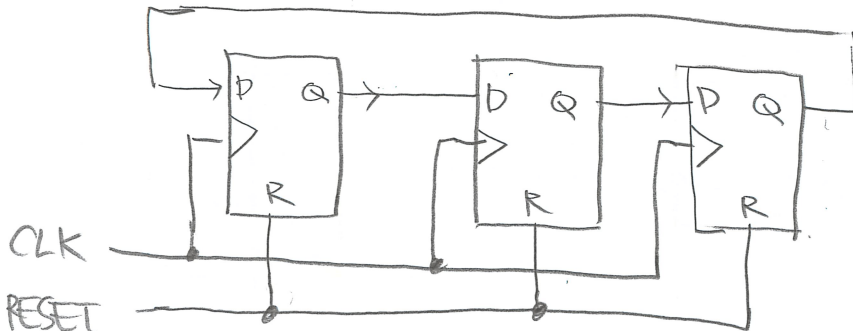
이렇게 구성한 000001 에서 100000 까지, 초기 시작은 100000 으로 시작한다.

그리고 표를 만들 때를 줄여서 나타내려면 2가지 방법이 있다.

① state 안에 output 정보 포함되므로 3바이트만 표현가능

② 디코더 사용하면 $\log_2 N$ 개로 줄일 수 있음. 대안하면 연산은 N input 값에 2^N output 이기 때문

① 3 바이트만 표현.



이때는 마지막 출력인 Q가 000 이고 처음 입력이 똑같이 000이기

때문에 바깥이 안쪽에 넣어주었다.

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."

6. A field programmable gate array (FPGA) uses look-up tables (LUTs) rather than logic gates to implement combinational logic. The ESAL NoGwaje 3 FPGA has propagation and contamination delays of 0.51 and 0.25 ns, respectively, for each LUT. It also contains flip-flops with propagation and contamination delays of 0.81 and 0.41 ns, and setup and hold times of 0.45 and 0 ns, respectively.

A. If you are building a logic circuit that needs to run at 130 MHz, how many consecutive LUTs can you use between two flip-flops? Assume there is no clock skew and no delay through wires between LUTs.

클럭 $T = \frac{1}{f} = \frac{1}{130 \times 10^6} = 7.69 \text{ ns} \approx 7.7 \text{ ns}$ (주 보정값 셋째 자리에서 반올림하여 둘째자리까지 표현하였음)

$$T \geq t_{pcq} + N \cdot t_{LUT} + t_{setup}$$

t_{pcq} : FF의 전파 지연 시간
 $N \cdot t_{LUT}$: 두 FF사이 바재개등한 개수 LUTs 개수
 t_{setup} : FF의 셋업 시간
 t_{LUT} : LUT의 전파 지연 시간

$$7.69 \text{ ns} \geq 0.81 \text{ ns} + N(0.51 \text{ ns}) + 0.45 \text{ ns}$$

$$6.43 \text{ ns} \geq N(0.51 \text{ ns})$$

$$12.61 \geq N \rightarrow \boxed{12}$$

B. Suppose that all paths between flip-flops pass through at least one LUT. How much clock skew can the FPGA have without violating the hold time?

(LUT) $t_{cd} : 0.25 \text{ ns}$

(FF) $t_{hold} : 0 \text{ ns}$

$t_{ccq} : 0.41 \text{ ns}$

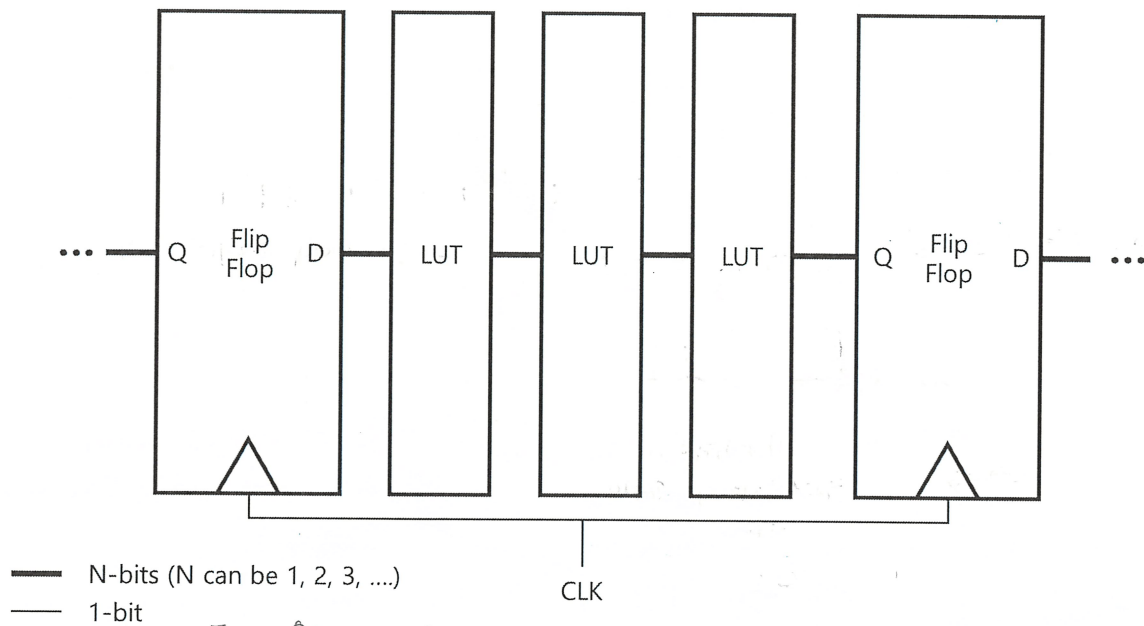
이 두 지연시간을 합치면 데이터가 FF ~ FF로 가장 빠르게 이동할 수 있는 시간임. 데이터가 너무 빨리 도착하여 hold time 조건을 위반하지 않도록 계산.

$$t_{skew} < (t_{ccq} + t_{cd}) - t_{hold} \quad \therefore t_{skew} < 0.41 \text{ ns} + 0.25 \text{ ns} - 0 \text{ ns} = \boxed{0.66 \text{ ns}}$$

→ 클럭 스레가 최소 하나의 LUT를 거치고, FF의 hold time을 위반하지

않는 최대값을 구하기 위한 공식이다. 이때, t_{ccq} 는 FF의 contamination delay를 의미하고, t_{cd} 는 LUT의 contamination delay를 의미한다.

- C. Assume a circuit which has three LUTs between two flip-flops. What is the maximum operating **frequency** of FPGA?



최대 운용주파수? 최소 지연을 갖는 설계일때임!

최종적으로, FF → FF로 데이터가 전송하는데 필요한 시간을 고려해 주기를 확보하고 이에 역수를 통해 구한다

이 원리에서 LUT는 세개가 직렬로 연결되어 있기 때문에 총 고려해야 하는것은 아래와 같다.

① 첫 번째 FF의 전파지연시간 : $t_{cq} = 0.81ns$

② LUT 자체의 전파지연시간 $\times 3$ (3개니까) : $t_{LUT} = 0.51ns$

③ 마지막 FF의 셋업시간 : $t_{setup} = 0.45ns$

∴ 총 지연시간 : $T = 0.81ns + 3 \times (0.51ns) + 0.45ns$
 $= 2.79ns$

∴ 최대 운용주파수

$$f = \frac{1}{T} = \frac{1}{2.79 \times 10^{-9}} = 0.358422 \times 10^9$$

$$= 358.42 \times 10^6$$

$$= \boxed{358.42MHz}$$

7. Compute the MTBF of a two-flipflop synchronizer and a three-flipflop synchronizer.

$$t_{\text{setup}} = t_{\text{hold}} = t_{\text{dCQ}} = t = 300\text{ps}$$

$$t_{\text{ck}} = 3\text{ns}$$

must sample a $f_{\text{input}} = 1\text{MHz}$ asynchronous signal

$$\tau = 100\text{ps} \text{ \& } G = 3$$

① 2FF synchronizer

$$\text{MTBF} = \frac{1}{R_F} = \frac{1}{f_{\text{input}} \cdot P_F} = \frac{1}{f_{\text{input}} \cdot P_E \cdot P_U}$$

$\frac{1}{R_F}$: Rate of Failure
 $f_{\text{input}} \cdot P_F$: Probability of Failure
 $f_{\text{input}} \cdot P_E \cdot P_U$: Probability of Uncertainty
 → 메타스테이블이 구간내에서 발생할 확률
 입력 D가 FF의 set up time & hold time 을 충족하지

$$\bullet f_{\text{input}} = 1\text{MHz}$$

$$\bullet P_E = \frac{t_{\text{setup}} + t_{\text{hold}}}{t_{\text{ck}}} = \frac{0.3\text{ns} + 0.3\text{ns}}{3\text{ns}} = 0.2$$

$$\bullet P_U = \frac{1}{G} e^{-\frac{t_w}{\tau}} = \frac{1}{3} e^{-\frac{2.4\text{ns}}{0.1\text{ns}}} = \frac{1}{3} \times e^{-24} = 1.258 \times 10^{-11}$$

$$t_w = t_{\text{ck}} - t_{\text{setup}} - t_{\text{hold}} = 3\text{ns} - 0.3\text{ns} - 0.3\text{ns} = 2.4\text{ns}$$

$$\circ \circ P_F = P_E \cdot P_U = 2.511 \times 10^{-12} / R_F = f_{\text{input}} \cdot P_F = 2.511 \times 10^{-6} [\text{sec}^{-1}]$$

$$\circ \circ \text{MTBF} = \frac{1}{R_F} = 397298 [\text{sec}]$$

② 3FF synchronizer

$$\bullet P_E = \frac{0.3\text{ns} + 0.3\text{ns}}{3\text{ns}} = 0.2$$

$$\bullet P_U = \frac{1}{3} e^{-\frac{4.8\text{ns}}{0.1\text{ns}}} = \frac{1}{3} e^{-48} = 4.7505 \times 10^{-22}$$

$$\bullet P_F = P_E \cdot P_U = 0.2 \times 4.7505 \times 10^{-22} = 9.501 \times 10^{-23}$$

$$\bullet R_F = 9.501 \times 10^{-17} [\text{s}^{-1}]$$

$$\bullet \text{MTBF} = \frac{1}{R_F} = 1.053 \times 10^{16} [\text{sec}]$$

"꿈을 가지십시오. 그리고 정열적이고 명예롭게 이루십시오."